

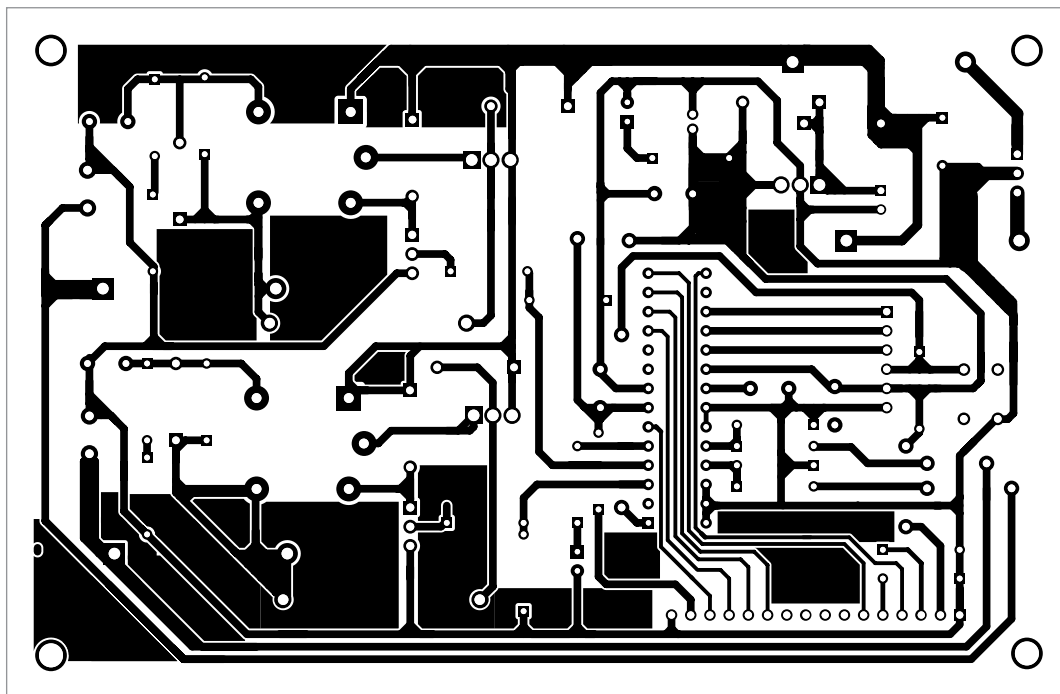


Fig. 2: Actual-size PCB layout for the cell charger

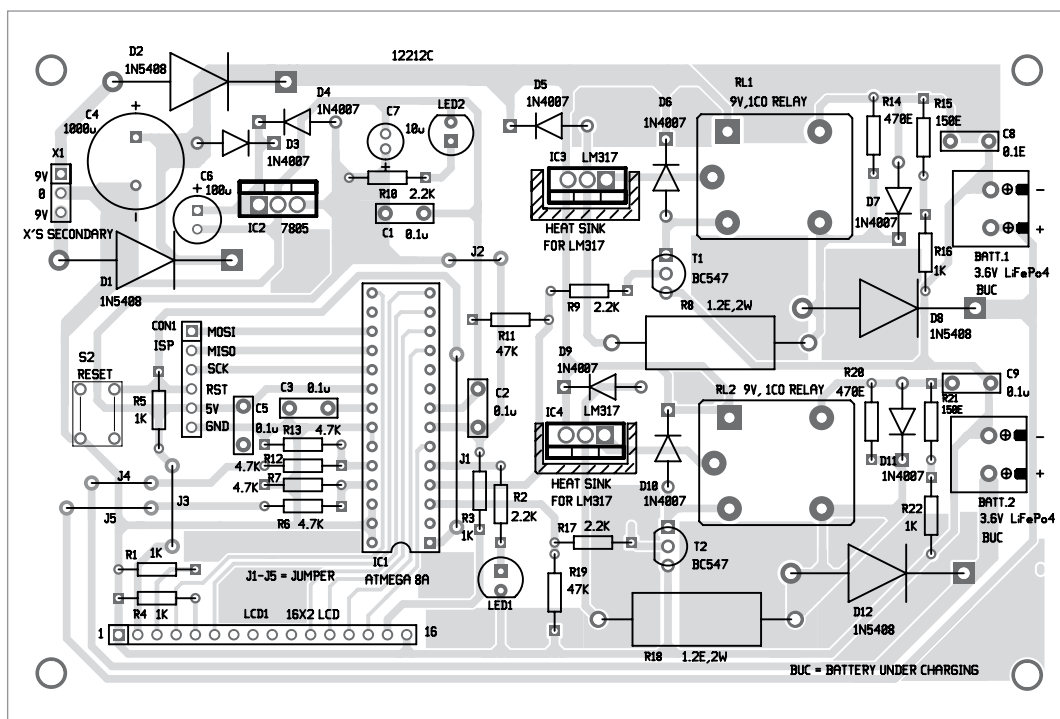


Fig. 3: Component layout of the PCB

(IC2), two LM317 adjustable voltage regulators (IC3 and IC4), a 16x2 LCD (LCD1), and two single-changeover relays (RL1 and RL2).

The charging system is controlled by ATmega8A microcontroller (MCU), which reads the charging voltage of

the cell continuously and changes the charging mode to either constant-current (CC) or constant-voltage (CV) accordingly. The same is displayed on the 16-column by 2-row text LCD.

The main concept of the charging system is based on LM317 voltage

regulator and the MCU is placed on the PCB. You can also use ISP port to write the hex code directly using in-system programming. All the functions and variables used in the code are named properly to identify their purpose during programming.

The voltage regulator may be wired in either constant-current or constant-voltage mode with small changes in the circuit. The changes in connection are achieved by controlling the relay's N/C and N/O contacts through the MCU. When the relay is on, the charger circuit is in CC mode, and 1A current is used for quick charging of the cell. Once the required cell voltage is achieved, the relay is switched off by the MCU and the charger circuit changes to CV mode.

The presence of the batteries, voltage of the batteries, status of the charging, mode of the charging (CC/CV), etc are all displayed on LCD1 for convenience.

Software

The program is developed in C language using AVR Studio4. It is compiled and the binary/hex code generated for the ATmega8A as *LiFePO4_charger.hex*.

Once the PCB is ready, the hex code is written/burnt to the MCU using a suitable AVR programmer and the MCU is placed on the PCB. You can also use ISP port to write the hex code directly using in-system programming. All the functions and variables used in the code are named properly to identify their purpose during programming.